

MT – Exercise 04

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1 Experiments with Byte Pair Encoding

Translation direction: Italian $\rightarrow$ English

use BPE	vocabulary size	BLEU
✗	2000	8.4
✓	2000	21.3
✓	4000	21.9

Manual inspection of the translations reveals substantial differences both across vocabulary sizes and between the use of BPE coding and no coding. Table 2 presents 7 example sentences selected from the test output. It is evident that the non-BPE-coded sentences do not produce adequate translations: they are frequently difficult to understand, and in the case of sentence 6, the output is lacking any and all linguistic information. Sentence 7 is missing what is arguably the single most important word in the sentence, the one that fundamentally determines the meaning.

Comparing the 2,000- and 4,000-vocabulary conditions, the results sometimes overlap, as is again the case with sentence 7; however, it is also to note that the larger vocabulary does not consistently improve translation quality. The BLEU scores corroborate this: a larger vocabulary does not necessarily yield better translations, at least under the current configuration.

The forked repo of the exercise can be found [here](#).

Table 1: MT output comparison per sentence.

#	Source	Word-Level	BPE—2000	BPE—4000	Reference
1	Ci ho messo un po' a capirlo.	I put it a little bit <unk>	I put a little bit to understand it.	I took a little bit to understand that.	And this took me awhile to figure out.
2	Una decina di anni fa ho scritto una storia sui vaccini per <i>The New Yorker</i> ; un breve articolo.	A <unk> <unk> I wrote a story about <unk> to <unk> New <unk> <unk>	A decades of years ago I wrote a story on vaccines for <i>The New Yorker</i> .	A decine years ago I wrote a story about the vaccines for <i>The New Yorker</i> ; a short article.	About a decade ago, I wrote a story about vaccines for <i>The New Yorker</i> . A little story.
3	Ed ero stupefatto che qualcuno si opponesse, che fosse contrario a quella che è, dopo tutto, la misura di salute pubblica più efficace di tutta la storia umana.	And I was <unk> that somebody <unk> that was <unk> to what is, after all, <unk> the measure of health <unk> more <unk> than all human history.	And I was amazed that somebody opposed to the one that it is, after all, the most effective public health measure of the human history.	And I was astonished that somebody opposed to what it is, after all, the way it is, after all, the most effective public health measure in human history.	And I was amazed to find opposition to what is, after all, the most effective public health measure in human history.
4	Non sapevo cosa fare, quindi ho fatto come al solito, ho scritto la storia e sono passato ad altro.	I didn't know what to do, so I did this <unk> I wrote the story and I wrote the story and I didn't know what to do.	I didn't know what to do, so I did as the sun, I wrote the story, and I went to the history.	And I didn't know what I do, so I did the usual I wrote the story and I went to something else.	I didn't know what to do, so I just did what I do: I wrote a story and I moved on.
5	Non molto più tardi, ho scritto una storia sul cibo geneticamente modificato.	Not much more <unk> I wrote a story about food.	Not much later, I wrote a story about food.	No much later, I wrote a story about genetically modified.	And soon after that, I wrote a story about genetically engineered food.
6	Stessa cosa, solo più grande.	<unk> <unk> <unk>	Same thing, just bigger.	It's just bigger.	Same thing, only bigger.
7	Le persone erano impazzite.	People were <unk>	People were crazy.	People were crazy.	People were going crazy.

2 Beam Search

Beam size	1	2	4	5	6	7	8	10	12	20
BLEU	19.7	21.4	21.9	21.9	21.8	21.8	21.8	21.8	21.7	21.4
time (s)	39	23	47	63	68	85	103	148	167	340

Table 2: Beam size vs. variable result values of the transformer 4k vocabulary model. The beam size of 4 presents as ideal, haven taken less than minute to compute with a BLEU score of 21.9.

Regarding beam size, a particularly interesting finding is that a larger beam does not automatically produce a higher BLEU score, as it does not always identify a better translation. The drop is slight—from a peak of 21.9 to 21.4—but a difference of this magnitude can meaningfully affect translation quality, as was already observed in a comparable example from Task 1. Translation time decreases in an approximately linear progression; the first run took slightly longer despite a beam size of one, which may be attributable to shell script initialisation overhead, but the overall difference remains small. That said, a beam size of two takes roughly double the time of beam size one in absolute terms.

The BLEU score peaks relatively quickly: beam size one yields noticeably poor translations compared to all other conditions, while beam sizes four and five—which was already the default in the example setup provided with this exercise—achieve the peak score of 21.9 at a runtime of 47 and 63 seconds respectively, which is comparatively low across the full range of beam sizes tested. For further translation tasks, a beam size of four or five is therefore recommended.